

Emily Alger

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I develop novel Bayesian adaptive dose-finding trial designs. My goal is to promote the adoption of rigorous Bayesian methods within clinical trial design and practice.

Education

October 2022 – October 2025: Institute of Cancer Research, PhD in Early Phase and Adaptive trials

- Awarded bursary of full fees and stipend for three years, supervised by Professor Christina Yap.

October 2017 – June 2022: University of Warwick, Mathematics and Statistics (BSc MMathStat)

- First Class - Bachelor of Science, Master of Mathematics and Statistics (with Honours).
- Relevant final year modules: Monte Carlo Methods (93), Statistical Learning and Big Data (82), Bayesian Statistics and Decision Theory (82), Bayesian Forecasting and Intervention (76).

Professional position

November 2025 - : University of Oxford, Postdoctoral Research Assistant

- Oxford-GSK Centre for Biostatistics and AI in Medicine, supervised by Professor Tom Rainforth.

Notable First Author Publications:

- Statistical Methods in Medical Research, 2026: [PRO-ADD: Patient-empowered dose-finding trials integrating safety, preliminary efficacy and patient-reported outcomes for optimal dose selection](#)
- Clinical Cancer Research, 2026: [Generalised pairwise comparisons in dose-optimisation oncology trials: beyond safety to multi-outcome dose selection](#)
- The Lancet Oncology, 2026: [A practical toolkit with recommendations for analysing and visualising patient-reported outcomes in early phase dose-finding oncology trials \(OPTIMISE-AR\)](#)
- Journal of Clinical Oncology, 2026: [International Consensus-Driven Recommendations for Patient-Reported Outcome Research Objectives in Early Phase Dose-Finding Oncology Trials: OPTIMISE-ROR](#)

Upcoming Statistical Methodology publications

Calibration of dose-agnostic priors for Bayesian dose-finding trial designs with joint outcomes

- Divergence minimisation to calibrate priors for noninformative a priori dose recommendations in joint-outcome dose-finding trials, including analytical derivations and extensive simulations.

Grants and Awards

- **May 2025: American Statistical Association Biopharmaceutical Section Scholarship Award**, recognising notable research and academic achievement within biopharmaceutical statistics.
- **May 2025: ESMO Merit Travel Grant**, Travel bursary to present at ESMO TAT Asia Congress 2025.
- **February 2025: Institute of Cancer Research Undergraduate Summer Scholarship Scheme**, secured funding to supervise an undergraduate summer student in summer 2025.
- **October 2024: TMRP Student Network Funding**, Travel grant to visit MD Anderson Cancer Centre in October 2024 to research adaptive borrowing techniques for Patient-Reported Outcomes.
- **May 2024: Student award, *Advanced Statistical Designs to Empower Biomarker-driven Clinical Trials* workshop**, Travel bursary to present at the University of Bath.
- **May 2024: Training Support Fund, The Alan Turing Institute**, Travel bursary to support my oral presentation at the Society for Clinical Trials annual general meeting in May 2024.
- **October 2023: Enrichment Scheme award, The Alan Turing Institute, London, UK**
- **March 2023: ESMO Merit Travel Grant**, Travel bursary to present at ESMO TAT 2023 in March 2023.
- **October 2022: Early-career researcher grant, 7th Early Phase Adaptive Trials Workshop.**
- **June 2022: Best Undergraduate Dissertation in Biostatistics, PSI.**
- **October 2021-June 2022: Statistics Senior Scholarship, University of Warwick.**

Research visits

October 2024: MD Anderson Cancer Centre, Houston, US.

- 10-day research visit to develop novel Phase I/II trial design to incorporate Patient-Reported Outcomes with Professor Jack Lee, Professor Ying Yuan and Dr Ruitao Lin.

May 2023: Columbia University, Mailman School of Public Health, New York, US.

- 3-week research visit to develop novel dose-finding trial designs which incorporate Patient-Reported Outcomes with Dr Shing Lee and Professor Ying Kuen Cheung.
- Associated paper published in [ESMO Open](#) with on-going collaboration on second project.

Talks and posters

- **Talk:** *Calibration of dose-agnostic priors for Bayesian dose-finding trial designs with joint outcomes*, 46th Annual Meeting of the Society for Clinical Trials.
- **Invited talk:** *The use of restricted mean survival time to estimate joint treatment effects in the presence of interaction terms under model misspecification - a simulation study*, Efficient study design discussion group, MRC Biostatistics Unit, University of Cambridge.
- **Talk:** *Patient-Centred Dose-Finding Trials using Safety, Efficacy and Patient-Reported Outcomes*, 45th Annual Conference of the International Society for Clinical Biostatistics.
- **Invited talk:** *Advanced Bayesian dose-finding adaptive designs for assessment of joint outcomes*, UCL Statistical Sciences PhD seminar, University College London.
- **Talk:** *Statistical methods and data visualisation of patient-reported outcomes in early phase dose-finding oncology trials: a methodological review*, 45th Annual Meeting of the Society for Clinical Trials.
- **Invited talk:** *U-PRO-CRM: Designing Patient-Centred Dose-Finding Trials with Patient-Reported Outcomes*, IDENT Research Dissemination Workshop, University of Bath.

Experience

October 2023 – July 2024: Enrichment Student, The Alan Turing Institute

- Established Bayesian Statistical Methods reading group: Generalised Bayes, Martingale posteriors.
- Collaborated with Academic services as an Enrichment student blogger.
- Successfully obtained funding to run student events to promote engagement and cohort building among the Turing community as part of my Community Champion role.

July – September 2022: Research Assistant, MRC Biostatistics Unit, University of Cambridge

- Investigated Restricted Mean Survival Time endpoint within clinical trial design.
- Simulated fixed sample and group sequential clinical trials in R using cluster computing techniques.
- Associated paper published in *Trials*, 2026: [Estimation of combined treatment effects by restricted mean survival time](#).

July – August 2021: Summer vacation student within Early Phase and Adaptive Trials, The Institute of Cancer Research

- Associated paper published in *Contemporary Clinical trials*: [Reporting quality of CONSORT flow diagrams in published early phase dose-finding clinical trial reports: Improvement is needed](#).

Services to the scientific community

- **Section Committee Member**, NIHR Statistics Group, Early Phase Clinical Trials.
- **Peer reviewer** for *Annals of Applied Statistics*, *Statistics in Medicine*, *Pharmaceutical Statistics*, *Nature Communications*, and *ESMO Open*.
- **On-site organiser** of the Trials Methodology and Research Partnership's Adaptive Design Working group in-person meeting, 16th-17th April 2024, hosted at the Institute of Cancer Research.